

Master Theorem Worksheet

This is a worksheet to help you master solving recurrence relations using the Master Theorem. For each recurrence, either give the asymptotic solution using the Master Theorem (state which case), or else state that the Master Theorem doesn't apply. You should be able to go through these **25** recurrences in **10** minutes.

Problem 1-1. $T(n) = 3T(n/2) + n^2$

$$n^{\log_2 3} \approx 1.585$$

$$n^{1.585} \rightarrow \Omega(n^2) \rightarrow \boxed{\text{case 3}}$$

Problem 1-2. $T(n) = 7T(n/2) + n^2$

$$n^{\log_2 7} = 2.807$$

$$n^{2.807} \rightarrow O(n^2) \rightarrow \boxed{\text{case 1}}$$

Problem 1-3. $T(n) = 4T(n/2) + n^2$

$$n^{\log_2 4} = 2$$

$$n^2 \rightarrow n^2 \rightarrow \boxed{\text{case 2}}$$

Problem 1-4. $T(n) = 3T(n/4) + n \lg n$

$$n^{\log_4 3} = 0.792481$$

$$n^{0.792481} < n \lg n \rightarrow \boxed{\text{case 3}}$$

Problem 1-5. $T(n) = 4T(n/2) + \lg n$

$$n^{\log_2 4} = n^2$$

$$n^2 > n \lg n \rightarrow \boxed{\text{case 1}}$$

Problem 1-6. $T(n) = T(n-1) + n$

Does not apply $\rightarrow$ no b

$$T(n) = aT(n/b) + f(n)$$

$$n^{\log_b a}$$

Case 1: $n = O(n^c)$
 for $c > 1$

Case 2: $n = \Theta(n(\log n)^k)$
 for $k \geq 0$

Case 3: $n = \Omega(n^c)$
 for $c > 1$

Problem 1-7. $T(n) = 4T(n/2) + n^2 \lg n$

$$n^{\log_4 2} = n^2$$

$$n^2 < n^2 \lg n \rightarrow \text{but only by scale of } \lg n \rightarrow \boxed{\text{case 2}}$$

Problem 1-8. $T(n) = 5T(n/2) + n^2 \lg n$

$$n^{\log_2 5} \approx n^{2.32}$$

$$n^{2.32} > n^2 \lg n \rightarrow \boxed{\text{case 1}}$$

Problem 1-9. $T(n) = 3T(n/3) + n/\lg n$

$$n^{\log_3 3} = n$$

$$n > n/\lg n \rightarrow \text{greater than by } \lg n \text{ but } k < 0 \rightarrow \boxed{\text{does not apply}}$$

Problem 1-10. $T(n) = 2T(n/4) + c$

$$n^{\log_4 2} = n^{1/2}$$

$$n^{1/2} > c \rightarrow \boxed{\text{case 1}}$$

Problem 1-11. $T(n) = T(n/4) + \lg n$

$$n^{\log_4 1} = 0$$

$$0 < \lg n \rightarrow \text{only by scale of } \lg n \text{ where } k > 0 \rightarrow \boxed{\text{case 2}}$$

Problem 1-12. $T(n) = T(n/2) + T(n/4) + n^2$

Multihop recursion doesn't work for master theorem

Problem 1-13. $T(n) = 2T(n/4) + \lg n$

$$n^{\log_4 2} = n^{1/2}$$

$$n^{1/2} > \lg n \rightarrow \boxed{\text{case 1}}$$

Problem 1-14. $T(n) = 3T(n/3) + n \lg n$

$$n^{\log_3 3} = n$$

$$n < n \lg n \text{ by scale of } \lg n \rightarrow \boxed{\text{case 2}}$$

Problem 1-15. $T(n) = 8T((n - \sqrt{n})/4) + n^2$

Master does not apply

Problem 1-16. $T(n) = 2T(n/4) + \sqrt{n}$

$$n^{\log_4 2} = n^{1/2}$$

$$n^{1/2} = n^{1/2} \rightarrow \boxed{\text{case 2}}$$

Problem 1-17. $T(n) = 2T(n/4) + n^{0.51}$

$$n^{\log_4 2} = n^{1/2}$$

$$n^{1/2} < n^{0.51} \rightarrow \boxed{\text{case 3}}$$

Problem 1-18. $T(n) = 16T(n/4) + n!$

$$n^{\log_4 16} = n^2$$

$$n^2 < n! \rightarrow \boxed{\text{case 3}}$$

Problem 1-19. $T(n) = 3T(n/2) + n$

$$n^{\log_2 3} = 1.585$$

$$n^{1.585} > n \rightarrow \boxed{\text{case 1}}$$

Problem 1-20. $T(n) = 4T(n/2) + cn$

$$n^{\log_2 4} = n^2$$

$$n^2 > cn \rightarrow \boxed{\text{case 1}}$$

Problem 1-21. $T(n) = 3T(n/3) + n/2$

$$n^{\log_3 3} = n$$

$$n > n/2 \rightarrow \text{not difference enough} \rightarrow \Theta(n \lg n) \rightarrow \boxed{\text{case 2}}$$

Problem 1-22. $T(n) = 4T(n/2) + n/\lg n$

$$n^{\log_2 4} = n^2$$

$$n^2 > n/\lg n \rightarrow \boxed{\text{case 1}}$$

Problem 1-23. $T(n) = 7T(n/3) + n^2$

$$n^{\log_3 7} \approx n^{1.77}$$

$$n^{1.77} < n^2 \rightarrow \boxed{\text{case 3}}$$

Problem 1-24. $T(n) = 8T(n/3) + 2^n$

$$n^{\log_3 8} = n^{1.892}$$

$$n^{1.892} < 2^n \rightarrow \boxed{\text{case 3}}$$

Problem 1-25. $T(n) = 16T(n/4) + n$

$$n^{\log_4 16} = n^2$$

$$n^2 > n \rightarrow \boxed{\text{case 1}}$$